

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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TITLE

Implementation of JavaScript Arrays, Functions, and Object-Oriented Programming (Classes)

AIM

To understand and implement JavaScript arrays and ES6 classes by building a student marks calculator and a library management system.

OBJECTIVES

- Create and manipulate arrays using standard JavaScript methods.
- Define and invoke functions to perform iterative calculations (e.g., total and average).
- Implement an ES6 class with a constructor and member methods.
- Manage and manipulate arrays of objects within a class structure.

TASK DESCRIPTION

Design two JavaScript programs. The first program must manage an array of student marks using functions to display the array, calculate the total, calculate the average, and append new marks. The second program must establish a Library class to manage a collection of book objects, including methods to add new books, display the catalog, calculate the total cost of all books, and remove the most recently added book.

ALGORITHM/PROCEDURE

1. Array & Functions: Define an array of numbers representing student marks.
2. Create reusable functions to display the array, loop through the array to calculate the total sum, calculate the mathematical average, and push a new mark to the array.
3. Execute these functions and log the outputs to the console.
4. Library Class: Define a Library class with a constructor that initializes an empty books array.
5. Implement class methods: addBook() to push new book objects (title and cost), displayBooks() to iterate and print the catalog, totalPrice() to sum the costs, and removeBook() to pop the last book from the array.
6. Instantiate the Library class, add several books, and execute the methods to verify functionality.

PROGRAM/SOURCE CODE

The following code represents the corrected and commented JavaScript programs:

```
// Functions.js

// 1. Initialize an array of student marks
let marks = [85, 90, 78, 88, 95];

// 2. Function to display the marks array
function displayMarks(arr) {
    console.log("Student Marks:", arr);
}

// 3. Function to calculate the total sum of marks
function totalMarks(arr) {
    let total = 0;
    // Iterate through the array and add each mark to the total
    for (let i = 0; i < arr.length; i++) {
        total += arr[i];
    }
    return total;
}

// 4. Function to calculate the average of the marks
function averageMarks(arr) {
    // Divide the total sum by the number of elements in the array
    return totalMarks(arr) / arr.length;
}

// 5. Function to append a new mark to the end of the array
function addMark(arr, mark) {
    arr.push(mark); // .push() adds the element to the array
    return arr;
}
```

```
// --- Execution & Logging ---
displayMarks(marks);
console.log("Total Marks:", totalMarks(marks));
console.log("Average Marks:", averageMarks(marks));

console.log("\nAfter Adding New Mark:");
addMark(marks, 92);
displayMarks(marks);
console.log("New Total:", totalMarks(marks));
```

```
// Object.js

console.log("==== Library Management System =====");

// 1. Define an ES6 Class named Library
class Library {
    // Constructor initializes an empty array to store book objects
    constructor() {
        this.books = [];
    }

    // 2. Method to add a new book object to the catalog
    addBook(name, price) {
        this.books.push({
            title: name,
            cost: price
        });
        console.log(name + " book added successfully.");
    }

    // 3. Method to iterate through and display all books
    displayBooks() {
        console.log("\nBooks in Library:");
        for (let i = 0; i < this.books.length; i++) {
            // Display 1-based index, title, and formatted cost
            console.log((i + 1) + ". " + this.books[i].title + " - ₹" +
this.books[i].cost);
        }
    }

    // 4. Method to calculate the total cost of all books in the array
    totalPrice() {
        let total = 0;
        for (let i = 0; i < this.books.length; i++) {
            total += this.books[i].cost;
        }
        return total;
    }
}
```

```

// 5. Method to remove the most recently added book (LIFO order)
removeBook() {
    // .pop() removes and returns the last element of the array
    let removedBook = this.books.pop();
    console.log("\nRemoved Book: " + removedBook.title);
}
}

// --- Instantiation and Testing ---
const lib = new Library(); // Create a new Library instance

// Adding books to the library
lib.addBook("Python", 500);
lib.addBook("Java", 600);
lib.addBook("HTML", 300);
lib.addBook("JavaScript", 400);

// Displaying the catalog and total price
lib.displayBooks();
console.log("\nTotal Price: ₹" + lib.totalPrice());

// Removing a book and displaying updated details
lib.removeBook();
console.log("\nAfter Removing One Book:");
lib.displayBooks();
console.log("\nNew Total Price: ₹" + lib.totalPrice());

```

CODE EXPLANATION

JavaScript Element	Purpose & Explanation
function	Defines reusable blocks of code utilized here for mathematical operations like calculating total and average marks.
push()	A built-in array method used to append a new element (a mark or a book object) to the end of an array.
class	ES6 syntax used to define a blueprint (Library) for creating objects with shared properties and methods.
constructor()	A special class method that automatically executes upon object instantiation to initialize the empty books array.
pop()	A built-in array method that removes the last element from an array and returns it, used here to remove the latest book.
for loop	Used to iterate over the length of the

	arrays to accumulate totals and print catalog lists dynamically.
--	--

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result	Status
TC-01	Array sum calculation	Sum of [85, 90, 78, 88, 95] equals 436	Output returns 436 correctly	Pass
TC-02	Array element addition	addMark() appends 92 to the array	Array length increases; new total is 528	Pass
TC-03	Object instantiation	lib.addBook() pushes objects into the class array	4 books are added and displayed with their respective costs	Pass
TC-04	Class logic execution	lib.totalPrice() returns the sum of 4 books	Total calculates to 1800	Pass
TC-05	Object removal via pop()	removeBook() deletes the last added object	"JavaScript" is removed; new total updates to 1400	Pass

OUTPUT

The following console output screenshots depict the execution of the 5 Test Cases based on the implemented JavaScript code:

TC-01: Array sum calculation

```
PS C:\Lab> node Functions.js
Student Marks: [ 85, 90, 78, 88, 95 ]
Total Marks: 436
```

TC-02: Array element addition

```
After Adding New Mark:  
Student Marks: [ 85, 90, 78, 88, 95, 92 ]  
New Total: 528
```

TC-03: Object instantiation and array push

```
PS C:\Lab> node Object.js  
===== Library Management System =====  
Python book added successfully.  
Java book added successfully.  
HTML book added successfully.  
JavaScript book added successfully.  
  
Books in Library:  
1. Python - ₹500  
2. Java - ₹600  
3. HTML - ₹300  
4. JavaScript - ₹400
```

TC-04: Class logic execution (Total Cost)

```
Total Price: ₹1800
```

TC-05: Object removal via pop()

```
Removed Book: JavaScript  
  
After Removing One Book:  
  
Books in Library:  
1. Python - ₹500  
2. Java - ₹600  
3. HTML - ₹300  
  
New Total Price: ₹1400
```

RESULT

Thus, JavaScript concepts involving basic array manipulation, function definitions, and Object-Oriented ES6 classes were successfully implemented and tested. The console

output verifies the accurate calculation of array logic and proper state management within the Library class.